

EDUCATION

Master of Science, Data Science

Aug 2025 – Aug 2026

University of Virginia · Charlottesville, VA

- Coursework: Statistical Learning · Deep Learning · Machine Learning · Big Data Systems · Data Pipeline Architecture · Data Visualization · Computation for Data Science · Understanding Uncertainty · ML Systems & Network Security · Ethics of Big Data.

Bachelor of Science, Game Design and Development

Aug 2019 – Dec 2024

Rochester Institute of Technology · Rochester, NY

- Minor in History. GPA: 3.5 / 4.0. Strong foundations in systems thinking, algorithmic design, and data-driven iteration.

PROJECTS

AI-Assisted Quality Monitoring for ALMA Proposal Reviews

2026

M.S. Capstone · 4-person team · Sponsored by the ALMA Observatory

- Built a prototype NLP and LLM pipeline to flag low-quality peer reviews for the Atacama Large Millimeter/submillimeter Array, which fields ~1,700 proposals and ~17,000 written reviews each annual cycle.
- Engineered and iteratively tuned an LLM scoring prompt rating reviews on rank-sentiment match, evaluative depth, professionalism, topical relevance, and abstract copying, calibrating thresholds to limit false positives.
- Compared classical NLP baselines (TF-IDF, astroBERT and MiniLM sentence embeddings, cosine similarity) against LLM scoring; delivered a documented pipeline in Dataiku usable by non-technical staff.

DuckDB vs. Spark Pipeline Benchmark

Summer 2026

Team Member

- Implemented parallel feature-engineering pipelines in DuckDB (SQL) and PySpark RDDs over 864 MB of NFL tracking data, computing per-play speed and route-distance features from raw positional frames.
- Benchmarked both across five data scales: DuckDB completed in 10.5s versus Spark's 218s, and verification confirmed identical outputs across all 14,251 plays; analyzed the memory and scaling trade-offs behind the gap.
- Extended the project with a Streamlit dashboard served from AWS for interactive player- and position-level exploration.

Fuel Economy Modeling and Vehicle Clustering

Winter 2025

Team Member

- Applied Ridge Regression, KNN, Logistic Regression, and K-Means with PCA to a large-scale EPA vehicle dataset; performed EDA, feature engineering, and cross-validation.
- Evaluated models using R^2 , RMSE, and ROC AUC; communicated findings through interactive dashboards designed for non-technical stakeholders.

Snake Evolution

Winter 2022

Co-Creator

- Implemented a neural network and genetic algorithm to train a population of Snake agents to improve over successive generations.
- Adjusted fitness functions and mutation parameters to refine agent behavior; tracked performance metrics to measure learning progress.

EXPERIENCE

Additive Manufacturing Engineer

Mar 2024 – May 2025

Architectural Graphics Inc. · Virginia Beach, VA

- Automated data-driven workflows using VBA and internal tools, cutting production engineer effort per project by 35%.
- Designed and optimized large-scale 3D printing pipelines, analyzing failure modes and bottlenecks to reduce lead times by ~25%.
- Developed and validated 50+ CAD models with built-in quality checks emphasizing repeatability and system-level optimization.

Frontend Web Developer Intern

May 2022 – Sep 2022

International STEM Academy · Atlanta, GA

- Built and deployed a data-informed responsive website serving 500+ monthly users, improving usability and accessibility.
- Collaborated with designers and stakeholders to iterate on features using engagement metrics, reducing bounce rate by 18%.

UX/UI Toolkit Developer Intern

Jun 2021 – Aug 2021

Rochester Institute of Technology · Rochester, NY

- Designed and implemented a reusable UI component library to support rapid prototyping and experimentation.
- Created 100+ widgets and 65 icons to enable consistent interface design across projects; adopted across multiple RIT courses.

SKILLS

Languages: Python, R, SQL, C#, JavaScript

Data & ML: pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers, sentence-transformers, tidyverse, ggplot2, matplotlib, seaborn, Jupyter

Big Data & Cloud: PySpark, DuckDB, AWS, Streamlit, Dataiku

ML Techniques: Regression, classification, clustering, PCA, deep learning, NLP and LLM prompt engineering, cross-validation, feature engineering, model evaluation

Tools & Platforms: Git, GitHub, VS Code, RStudio, Tableau, Excel, Google Colab, Firestore

Web & Other: React, Node.js, HTML/CSS, Angular, VBA, AutoCAD, Unity, Godot